

HW02:

Problem 1) a

$$3n^2 - 8n + 4 \quad \text{upper bound}$$

$$3n^2 + 4n^2 = 7n^2$$

lower bound

$$3n^2 - 8n$$

$$n^2 + 2n^2 = 3n^2$$

$$2n^2 + n^2 = 3n^2 \geq 8n$$

$$2n^2$$

$$n \geq 8$$

$$C_1 = 2$$

$$C_2 = 7$$

$$C_1 g(n) \leq f(n) \leq C_2 g(n)$$

$$f(n) = \Theta(n^2)$$

$$b) \frac{n^2 + 2n - 5}{n - 10}$$

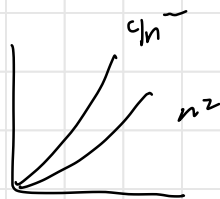
$$\text{upper bound} = \frac{\text{upper}}{\text{lower}}$$

$$\text{Lower bound} = \frac{\text{Lower}}{\text{upper}}$$

$$n^2 + 2n - 5 \leq n^2 \leq n^2 + 2n = n^2 + 2n^2 = 3n^2 \quad \frac{3n^2}{n^2} = 3$$

$$n - 10 \geq \frac{1}{2}n \quad C = \frac{1}{2} = \frac{1}{2} \quad \frac{n^2 + 2n - 5}{n - 10} \geq \frac{1}{2}n^2 \quad C_2 = 1 \quad \frac{n^2}{n - 10} \geq \frac{1}{2}n^2 \quad \frac{n^2}{n} = n$$

$$c) f(n) = \begin{cases} n^2, & n \text{ is odd} \\ 4n^2, & n \text{ is even} \end{cases}$$



$$C_1 = 1$$

$$C_2 = 4$$

$$C_1 g(n) \leq f(n) \leq C_2 g(n)$$

$$f(n) = \Theta(n^2)$$

upper bound

$$\text{odd} \quad n^2 \leq n^2 \leq 4n^2$$

$$\text{even} \quad 4n^2 \leq 4n^2$$

$$\therefore C_2 = 4$$

$$C_1 g(n) \leq f(n) \leq C_2 g(n)$$

$$f(n) = \Theta(n^2)$$

lower bound

$$\text{odd} \quad n^2 \leq n^2$$

$$\text{even} \quad n^2 \leq 4n^2 \leq 4n^2 \quad \therefore C_1 = 1$$

$$d) f(n) = \frac{n^3 - n^2 + n + 1000}{(n-1)(n+2)} \leq \frac{\text{upper bound}}{(n-1)(n+2)}$$

$$\leq \frac{n^3 + n^3 + 1000n^3}{(n-1)(n+2)} = \frac{102n^3}{(n-1)(n+2)} = \frac{102n^3}{\frac{1}{2}n^2} = 204n$$

$$n^3 - n^2 + n + 1000 \leq \frac{1}{2}n^3 \quad \text{for } n \geq 1000$$

$$\text{lower bound} \quad C_1 n^2 \leq f(n)$$

$$C_2 = 51$$

$$f(n) \leq 51n$$

$$\frac{n^3 - n^2 + n + 1000}{(n-1)(n+2)} = \frac{n^3 - n^2}{(n-1)(n+2)}$$

$$= \frac{n^3 - n^2}{(n-1)(n+2)}$$

$$-n^2 \leq \frac{1}{2}n^3$$

$$n-1 \leq n$$

$$n+2 \leq 2n$$

$$\text{for } n \geq 2$$

$$\geq \frac{\frac{1}{2}n^3}{n(n+2)}$$

$$= \frac{\frac{1}{2}n^3}{2n^2}$$

$$= \frac{1}{4}n$$

$$C_1 = \frac{1}{4}$$

$$f(n) \geq \frac{1}{4}n \quad n \geq 2$$

Problem 2 a) $T_1(n) + T_5(n) = \Theta(n^{2.5}) + \Theta(n) = \Theta(n^{2.5})$

b) $T_2(n) + T_6(n) = O(n \log n) + \Theta(n \log n) = \Theta(n \log n)$

c) $T_2(n) + T_4(n) = O(n \log n) + (O(n^4) + \Omega(n^2)) = O(n^4) \neq \Omega(n^2)$

d) $T_7(n) + T_4(n) = O(n^{1.5} \log n) + \Omega(n \log n) + O(n^4) + \Omega(n^2) = O(n^4) \neq \Omega(n^2)$

e) $T_3(n) + T_1(n) = \Omega(\log n) + \Theta(n^{2.5}) = \Omega(n^{2.5})$

f) $T_1(n) + T_4(n) = \Theta(n^{2.5}) \times (O(n^4) \neq \Omega(n^2)) = O(n^{6.5})$
 $\Omega(n^{4.5})$

g) $T_6(n) + \frac{T_4(n)}{T_5(n)} = \Theta(n \log n) + \frac{O(n^4) \neq \Omega(n^2)}{\Theta(n)}$

$\Theta(n \log n) + O(n^3) \Omega(n) = O(n^3) \neq \Omega(n \log n)$

Problem 3 a) best case $n \rightarrow$ when during the first outer loop and after the first inner the $bool = true$ and the function returns

worst case $n^2 \rightarrow$ if k is at the end of the array

b) best case $n \rightarrow$ when part a is best cause and then the linear search finds the element on the first try

worst case $n^2 + n = n^2 \rightarrow$ 1st largest worst case is n^2 then linear search is n worst case where the number is at the end of the array

Problem 4 a)

$$T_{avg}(n) = \sum_{\text{case \& all case}} p(\text{case}) \cdot T(\text{case})$$

$$\text{Case 1 both heads } 0.5 \cdot 0.5 = .25 \quad T(\text{Case 1}) = \Theta(n)$$

$$\text{Case 2 one coin not head } : .75 \quad T(\text{Case 2}) = \Theta(n^2)$$

$$T_{avg}(n) = 0.25 \cdot \Theta(n) + 0.75 \cdot \Theta(n^2) \therefore \Theta(n^2)$$

b $T_{avg}(n) = \sum p(\text{case}) \cdot T(\text{case})$

$$\text{Case 1 } k < np \cdot \sqrt{n} \quad p(\text{Case 1}) = \frac{1}{\sqrt{n}} \quad T(\text{Case 1}) = \Theta(n^2)$$

$$\text{Case 2 } k > np \cdot \sqrt{n} \quad p(\text{Case 2}) = 1 - \frac{1}{\sqrt{n}} \quad T(\text{Case 2}) = \Theta(1)$$

$$T_{avg}(n) = \frac{1}{\sqrt{n}} \cdot \Theta(n^2) + (1 - \frac{1}{\sqrt{n}}) \cdot \Theta(1) = \Theta(n^{-0.5}) \cdot \Theta(n^2) = \Theta(n^{1.5})$$

c $T_{avg}(n) = \sum p(\text{case}) \cdot T(\text{case})$

$$\text{Case 1 : get a number } p(\text{Case 1}) = \frac{1}{n} \quad T(\text{Case 1}) = \frac{n(n+1)(2n+1)}{6}$$

$$= \Theta(n^3)$$

$$T_{avg}(n) = \frac{1}{n} \cdot \Theta(n^3) = \Theta(n^2)$$

Problem 5 a $\Omega(n)$, because you can have a list alternating T & F and you would need to count all items in the list to see which one appears more frequently

b $\Omega(n \log n)$ w/ mergesort we reviewed in lecture has the lower bound is $\Omega(n \log n)$

c $\Omega(\sqrt{n})$ because we know the largest values for each column is at the end so that lookup is $\Theta(1)$ and there will be $\sqrt{n}$ rows to search through for the max